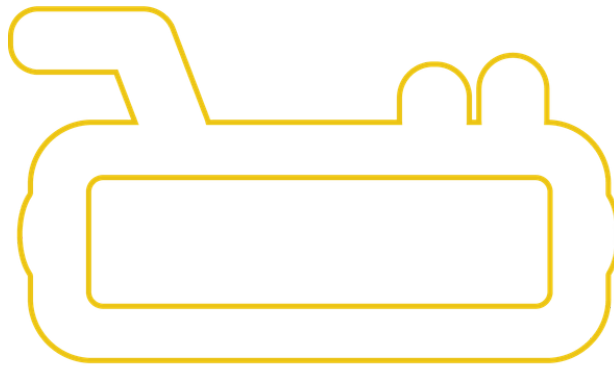
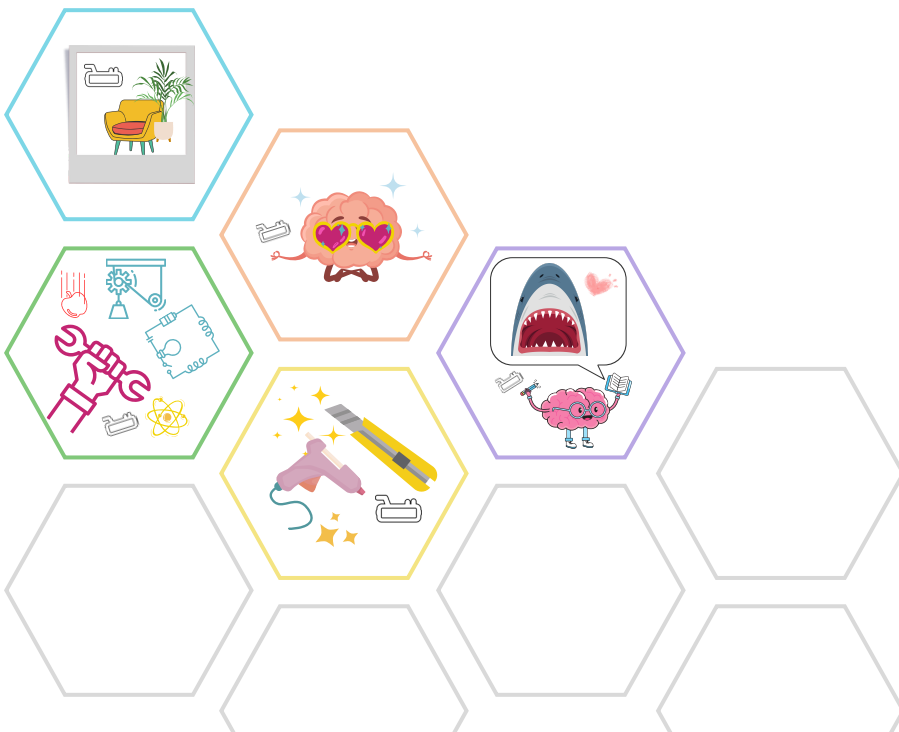




STEAMHEAD
MAKE • GREAT • THINGS

STEAMHEAD MAKES!

MAKER _____



Hexagons # 26 - 50

HEXAGON

#1

Skill

Maker Mindset: Learn from Failures



Activity

REFLECTION

Hexagon #

Date

I learned...

I want to know more about...

Notes / Feedback



HEXAGON

#1

Skill

Maker Mindset: Learn from Failures



Activity

REFLECTION

Hexagon #

Date

I learned...

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Notes / Feedback

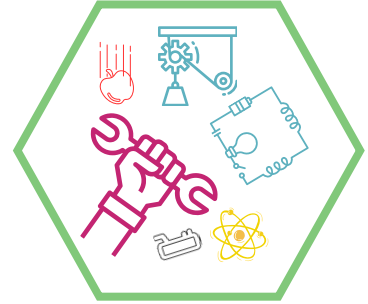


HEXAGON

#1

Skill

Maker Mindset: Learn from Failures



Activity

REFLECTION

Hexagon #

Date

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I want to know more about...

Notes / Feedback

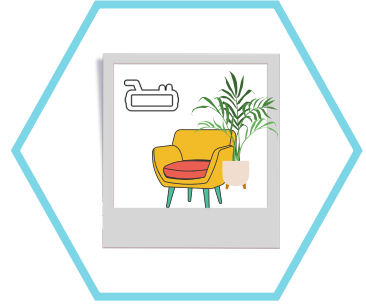


HEXAGON

#1

Skill

Maker Mindset: Learn from Failures



Activity

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Notes / Feedback

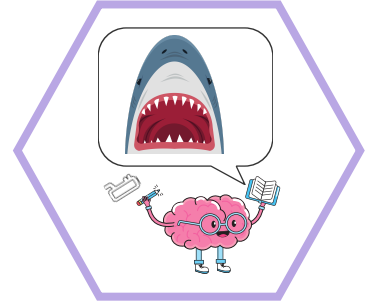


HEXAGON

#1

Skill

Maker Mindset: Learn from Failures



Activity

REFLECTION

Hexagon #

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Notes / Feedback

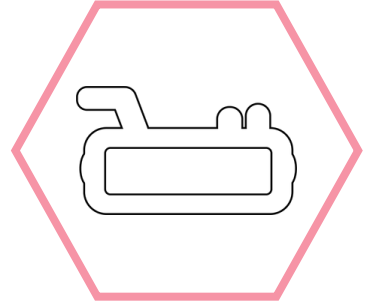


HEXAGON

#1

Skill

Maker Mindset: Learn from Failures



Activity



REFLECTION

Hexagon #

Date

I learned...

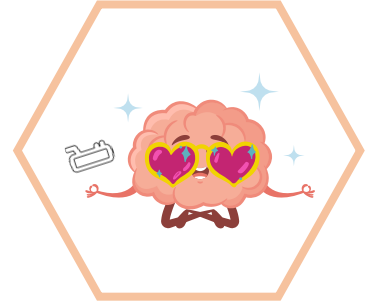
I want to know more about...

Notes / Feedback



Skill

Maker Mindset: Learns from Peer Feedback



Activity

I received feedback on one of my ideas or projects.

Who gave you the feedback?

What was the feedback about?

Circle how you felt about the feedback:



How did you respond or make changes based on the feedback?

I won't make the same mistake next time because I will...

How can feedback help improve your work in the makerspace?

Skill

Maker Mindset: Deeper Collaboration



Activity

I worked on a group project today.

Who were the members of your group?

How did you help your group?

Were there any disagreements? If yes, how did you resolve them?

Was it easy to work in a group? What could make it better?

Skill

Maker Mindset: Advanced Time Management



Activity

I planned my day and made a task checklist.

List three tasks you wanted to accomplish today.

Which tasks did you start? Which ones of those did you finish?

Which tasks did you postpone?

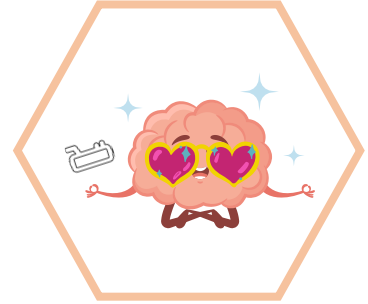
If you postponed a task, what was the reason?

Did you like creating your own list?

Should we do this more in the future?

Skill

Maker Mindset: Fostering a Positive Environment



Activity

I encouraged or supported a teammate.

Who did you support or encourage today?

How did you support them? (e.g., gave them a compliment, helped with a task, listened to their ideas)

How did it make them feel? (Ask them and write down their response.)

Why is it important to foster a positive environment in the makerspace?

Skill

Maker Mindset: Embracing New Challenges



Activity

I tried something new or challenging today.

What new or challenging task did you try?

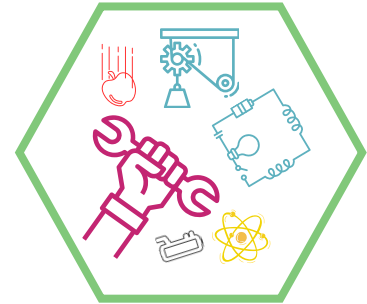
Were you successful? Why or why not?

If you faced difficulties, who or what helped you push through?

Why is it important to embrace new challenges in the makerspace?

Skill

Engineering Skills: Flashing Heart



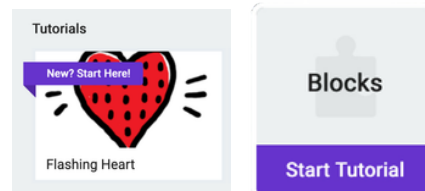
Activity

Know what a micro:bit is, how to get it started, and make it do something!

Goal: Create a flashing heart on the micro:bit LED display in multiple ways

Materials:

- micro:bit
- Computer
- Makecode.microbit.org



Do:

1. Open makecode.microbit.org on your computer, and find the **Flashing Heart** Tutorial. Click on it then **Blocks**.
2. Try out the tutorial!
3. If you have a micro:bit device, connect it to your computer and click the Download button. Follow the instructions to transfer your code onto the micro:bit and watch the hearts flash!
4. Did it work? Congratulations you coded instructions for a computer!

If it did not work, try again or ask a friend for help. No worries! If you have time, design your own shape and make it flash.

REFLECTION

Hexagon #

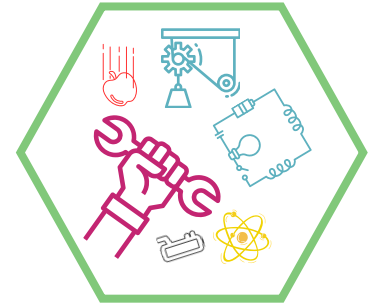
Date

Did you do anything to help a friend? If yes, how did you help them?

How did you feel about this project right when you opened the website?

How did you feel after you made your design flash?





Skill

Engineering Skills: Control the Flash

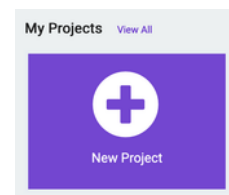
Activity

Control the flashing of your design in different ways

Goal: Create a microbit LED design that turns on and off at your command.

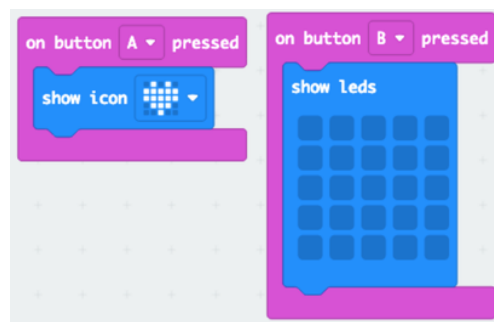
Materials:

- micro:bit
- Computer with block editor



Do:

1. Open makecode.microbit.org on your computer, and find the **New Project** Tutorial.
2. Try out this code, modify it with your own two designs.
3. Replace the "on button" blocks with "on shake" or "on tilt".
4. Connect your micro:bit to a battery, then create a dance that changes between the two designs



REFLECTION

Hexagon #

Date

Did you help anyone, and did anyone help you? On what part?

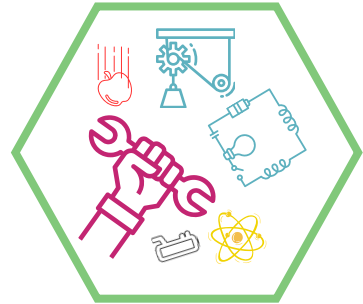
I want to know more about...

Notes / Feedback



Skill

Engineering Skills: Random Numbers



Activity

Let's create some digital dice!

Goal: Create a microbit that shows a new random number depending on what you do.

Materials:

- micro:bit
- Computer with block editor



Do:

1. Open makecode.microbit.org on your computer, and find the **Dice** Tutorial. Click on it then **Blocks**.
2. Try out the tutorial! If you have a micro:bit, click **Download** and follow the instructions.
3. Hack the Dice program to work when you press button A.

What other *Inputs* can be used to start the program?

REFLECTION

Hexagon #

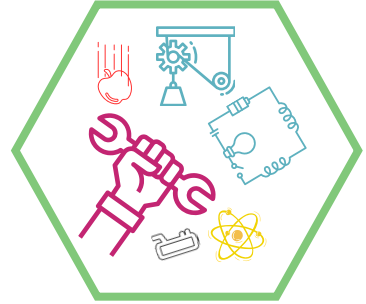
Date

How did you find out where the button A code block was? Did you help anyone else find it?

How could this be used to make something useful in a classroom, maybe to help decide something, what could it be useful for?

Notes / Feedback



Skill**Engineering Skills: micro:bit Counter****Activity**

Create a counter on the LEDs, using the advanced tutorial webpage.

Materials:

- micro:bit
- Computer with block editor

Do:

1. Visit **makecode.microbit.org/projects/step-counter** and complete the tutorial.
2. Test out your counter! Walk in place or slowly around the room
3. **Stretch goal:** In addition to testing the “on shake” function, hack the program and add “on button A” for increasing the counter.

REFLECTION

Hexagon #

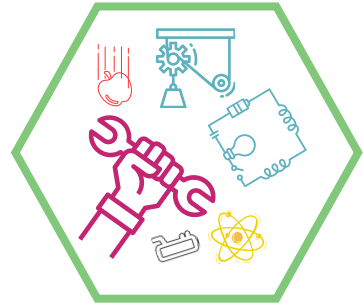
Date

Imagine and describe a way you could use counters in a game?

Did you do anything to help a friend? If yes, how did you help them?

How could you learn more micro:bit without a teacher?



**Skill****Engineering Skills: micro:bit Touch Sensor****Activity****Create a counter that displays on the LEDs, but without a tutorial!****Materials:**

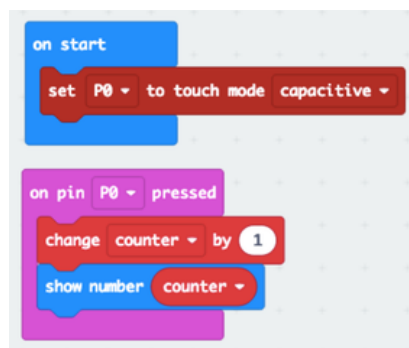
- micro:bit
- Computer with block editor

Do:

1. Find this block at **Advanced/Pins/More**.



2. Create the code below, upload it, and count upwards by touching the number 1 on your micro:bit.



3. Add an alligator clip to pin 1, and touch the end of it. What if you hold one end, and your friend touches you? Test the limits of the touch sensor.

REFLECTION

Hexagon #

Date

Imagine and describe a way you could use counters in a game?

Did you do anything to help a friend? If yes, how did you help them?

How could you learn more micro:bit without a teacher?

